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Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Sea Surface Temperature over the Black Sea

Issue: 5.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/10/09	All	Creation of the documents	Andrea Pisano, Bruno Buongiorno Nardelli	
2.0	2019/02/20	All	Reviewed document	Andrea Pisano, Bruno Buongiorno Nardelli	
3.0	2020/04/14		OMI update to 2018 and updated references	Andrea Pisano, Bruno Buongiorno Nardelli	
3.1	2020/09/10		Update of the upstream reference dataset and OMI extension to 2019	Andrea Pisano, Bruno Buongiorno Nardelli	
3.2	2021/12/14		OMI extension to 2020	Andrea Pisano, Claudia Fanelli, Bruno Buongiorno Nardelli	
4.0	2022/06/16		Addition of the year 2021 + automatic temporal extension	Andrea Pisano, Claudia Fanelli, Claudia Cesarini, Flavio La Padula, Bruno Buongiorno	A. Pisano

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5.0	2024/05/31	All	Update for 2023; Change of climatology to 1991-2020; Backward extension to 1982	Andrea Pisano, Claudia Fanelli, Claudia Cesarini, Flavio La Padula, Bruno Buongiorno	A. Pisano

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I INTRODUCTION

I.1 Product name

blksea_omi_tempsal_sst_area_averaged_anomalies

This OMI product includes unfiltered SST anomaly, given as monthly mean time series starting on 1982 and averaged over the Black Sea, and filtered SST anomaly (Tt), obtained by using the X11-seasonal adjustment procedure. Anomalies are computed against the 1991-2020 reference period.

blksea_omi_tempsal_sst_trend

This OMI product includes the cumulative/net trend map for the Black Sea, i.e. the rate of change of SST (°C/year) scaled by the number of time steps (namely, the number of years). Trend computation is based on X-11 seasonal adjustment procedure.

I.2 Indicator evaluation method

The Black Sea (BLKSEA) Sea Surface Temperature (SST) Ocean Monitoring Indicators (OMIs) are derived from the CMEMS Reprocessed Black Sea L4 (BS-REP-L4) SST product and its update runs in parallel with the corresponding BS-REP-L4 update. The BS-REP-L4 product provides the SSTs that are used to compute all the Black Sea SST OMIs, i.e. the time series of SST anomalies (unfiltered and filtered, blksea_omi_sst_area_averaged_anomalies) and the 2D map of SST trend (blksea_omi_sst_trend). Anomalies are computed against the 1991-2020 reference period. The 30-year climatology 1991-2020 is defined according to the WMO recommendation (WMO, 2017) and recent U.S. National Oceanic and Atmospheric Administration practice (https://wmo.int/media/news/updated-30-year-reference-period-reflects-changing-climate). These indicators belong to the ocean temperature and salinity (TEMPSAL) OMI family.

In detail, daily maps of BS-REP-L4 data from 1982 were averaged to obtain SST monthly maps, which were then used to build the time series of monthly mean SST values averaged over the Black Sea basin.

Trend analysis has been performed by using the X-11 seasonal adjustment procedure (Pezzulli et al. 2005). This procedure decomposes the input SST time series into the sum of three distinct components:

$$SSTt = Tt + St + It,$$

where Tt is the trend component, St and It the seasonal and the irregular components. The X-11 procedure has the effect of filtering the input time series with Tt acting as a low bandpass filter for interannual variations. The only difference with respect to the standard X-11 procedure is that we substituted the 13-term moving-average window (used to estimate Tt) with a 25-term moving-average window, i.e. from 1-year to 2-year low-pass filter. This makes more robust the estimation of the trend component Tt.

Mann-Kendall test (Mann, 1945; Kendall, 1975) was applied to Tt in order to assess whether there is a monotonic upward or downward trend over time and Sens's method (Sen, 1968) was subsequently applied to estimate the slope (°C/year) of the trend component Tt and its 95% confidence interval.

Finally, these OMIs were used and discussed in the first and second CMEMS Ocean State Report (Roquet et al., 2016; Mulet et al., 2018).

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
SST_BS_SST_L4_REP_OBSERVATIONS_010_022	QUID: http://marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-021-022-041-042.pdf PUM: http://marine.copernicus.eu/documents/PUM/CMEMS-SST-PUM-010-021-022-041-042.pdf

II SCIENTIFIC VALIDATION

The reliability of the BLKSEA SST OMI products is strictly related to the quality and accuracy of the BS-REP-L4 SST product, which is validated in the corresponding QUID (see http://marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-021-022-041-042.pdf). The accuracy of the BS-REP-L4 product has been assessed by comparison to independent in situ data for the period 2005-2018. Results show an overall mean bias and root mean square difference (RMSD) of the differences between REP BS and in situ data of 0.00 ± 0.01 K and 0.50 ± 0.01 K, respectively. The validation also shows a good stability of this dataset over the years, indicating that this product represents a valuable SST dataset for climate studies.

We recall that the BS-REP-L4 product is a collection of daily (nighttime) optimally interpolated (level-4, L4) satellite-based estimates of the foundation SST over the Black Sea at 0.05° grid presently covering the period 1982-present. This product is built from a consistent reprocessing of the level-3 (merged multi-sensor, L3) climate data record provided by the ESA Climate Change Initiative (CCI) (Merchant et al., 2019) and Copernicus Climate Change Service (C3S) initiatives, but also include in input an adjusted version of the AVHRR Pathfinder dataset version 5.3 (Saha et al., 2018) to increase the input observation coverage.

III REFERENCES

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